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Whose Voice Is It Anyway? Understanding AI Customization and Responsibility Attribution in Human-AI Collaboration

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ABSTRACT

This research examines how AI customization influences psychological ownership and responsibility attribution in human-AI collaboration. Through an online experiment with 507 participants using an AI assistant (ChatGPT 4o-mini), we found that customization enhances psychological ownership of the AI assistant that later transfers to ownership of the AI assistant's work. This process leads to higher satisfaction that persists even after receiving negative reviews on the output. Additionally, we found that participants who customized their AI attributed greater personal responsibility for negative outcomes. Interestingly, this effect was moderated by reviewer identity (a human expert reviewer, a human non-expert reviewer, and an AI reviewer). Notably, participants treated AI reviewers similarly to human experts in terms of perceived expertise and responsibility attribution, viewing both as more legitimate feedback sources than human non-experts. While customization can enhance human-AI collaboration through psychological ownership, it also reveals potential risks where individuals might create echo chambers of validation, potentially compromising critical evaluation of AI-generated work. This study is among the first to examine the dual pathway through which customization influences psychological ownership of both AI systems and their outputs, and uniquely demonstrates that AI reviewers are perceived as equivalent to human experts in evaluative contexts.

KEYWORDS

AI customization;
psychological ownership;
responsibility attribution; AI
credibility; Human-AI
collaboration

1. Introduction

The emergence of sophisticated artificial intelligence (AI) systems has fundamentally transformed how humans engage in cognitive and creative work (Dwivedi et al., 2021; Lee et al., 2022; Makridakis, 2017). It is projected that AI will have a cumulative global economic impact of \$19.9 trillion from 2024 to 2030 (International Data Corporation, October 2024), highlighting the transformative impact of these technologies on everyday life. Generative AI is capable of creating new content (e.g., text, images, music, videos, codes) through learned patterns from existing data (Fui-Hoon Nah et al., 2023). These generative AI systems assist individuals with varying levels of domain expertise (Pinski & Benlian, 2024) to approach and perform various types of tasks such as technical writing (Chubb et al., 2022; Wagner et al., 2022), creative production (Chakrabarty et al., 2024; Hitsuwari et al., 2023; Lima et al., 2021), individual- and organization-level decision-making (Jarrahi, 2018; Kim et al., 2021), and providing business solutions (Tarafdar et al., 2019; Vial et al., 2023). Unlike traditional automation tools that simply execute predefined tasks, modern AI systems—particularly large language models (LLMs)—can engage in dynamic dialogue, adapt to user preferences, and contribute meaningfully to complex cognitive tasks (Jo, 2023; Raisch & Krakowski, 2021), such as ChatGPT and Claude. This transformation positions AI not just as a tool but as a potential collaborator, and creates new challenges for understanding human-AI interaction, as user satisfaction, engagement, and perceived responsibility in these

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collaborative relationships can play significant role in the success of AI integration in both work settings and everyday activities (Binz et al., 2025). While technical capabilities of AI systems continue to advance, the psychological dimensions of how humans perceive, relate to, and collaborate with AI remain poorly understood (Stahl et al., 2021).

To delve deeper into the specific ways AI systems can transform individual users' workflows and foster collaboration, we now focus on LLMs, whose unique capabilities have reshaped the nature of human-AI interactions. The impact of LLMs on professional work is particularly significant because they have the capacity to serve as interactive collaborators rather than mere tools (Burton et al., 2024; Fui-Hoon Nah et al., 2023). Unlike traditional software that follows rigid rules, these systems can engage in dynamic dialogue, adapt to user preferences, and contribute meaningfully to complex cognitive tasks. This capability has led to their rapid adoption across various industries and domains. However, despite the versatility of AI systems, their effectiveness in human-AI collaboration is not uniform across individual users (Binz et al., 2025). For instance, generic AI responses may not always align with individual users' specific needs, communication styles, or decision-making preferences, potentially leading to frustration or inefficiencies in collaboration (Baek et al., 2025). Without mechanisms for personalization or customization, users may struggle to develop a sense of control and ownership over AI-assisted work, reducing their engagement and satisfaction in AI-generated outputs (Schwartz et al., 2023). Given these challenges, customization plays a crucial role in ensuring that AI systems are not just intelligent collaborators but also adaptable to users' unique workflows and expectations. Our study builds the theoretical framework by integrating psychological ownership theory (Pierce et al., 2001, 2003), attribution theory (Heider, 1958; Kelley, 1973), and source credibility theory (Pornpitakpan, 2004), and extending them to the novel and complex domain of human-AI collaboration to understand how customization can help attune AI systems to individuals' needs and preferences in the collaboration process.

2. Review of literature and theoretical underpinnings

2.1. The role of customization in human-AI collaboration

The importance of customization has been long established in the human-computer interaction literature. Early work on "tailorability" (Mackay, 1990) and the more recent work on "end-user development" (Barricelli et al., 2019) demonstrated that allowing individuals to adapt systems to their needs fosters greater satisfaction and better interactions. With LLMs, customization primarily manifests in two different ways: (1) output-style customization, where users can specify their preferred format, style, or content of the model's responses, and (2) output-goal customization, where users can influence how the model processes information and makes decisions (Gao & Liu, 2023). These customization capabilities are critical for effective human-AI collaboration because they enable users to align AI outputs with their specific communication needs, professional contexts, and task requirements. When users can customize an AI assistant's response style to match their organization's tone or technical vocabulary, the resulting outputs often require less editing and are more immediately useful, thus significantly increasing work efficiency. Similarly, when users can influence how an AI assistant processes information, they can guide it toward reasoning approaches that complement their thinking processes, creating a more synergistic collaborative relationship. This important adaptability transforms AI assistants' role from the traditional, generic "tool" into a more personalized collaborator that can extend and enhance users' unique capabilities and work styles.

The shift toward AI as a collaborative partner introduces unique considerations for customization. While traditional software customization focused primarily on interface modifications, AI customization encompasses how the system communicates, reasons, and collaborates with individuals. Guidelines for human-AI interaction emphasize the importance of giving individuals meaningful control over how AI behaves and communicates (Amershi et al., 2019; Lee & See, 2004), as this "co-adaptive" relationship strengthens trust and fosters a sense of partnership. However, customization in AI can present both opportunities and challenges. While it can enhance user engagement and reduce cognitive load (Liu, 2021), excessive customization options can lead to confusion or decision fatigue (e.g., individual users

may be overwhelmed when they are presented with numerous options and parameters to adjust, Amershi et al., 2019). Additionally, Nardi (1993) suggested that users who have different technical expertise may engage customization tools in different ways. When customization tools become too complex for individual users to understand, they may become overwhelmed, leading to reduced adoption and diminished benefits. This challenge can be particularly relevant to the “black box” nature of AI (Lipton, 2018; Ribeiro et al., 2016), where the underlying mechanisms of how these models process information and generate responses remain unclear to researchers and users. In the context of human-AI collaboration, the “black box” nature creates an additional layer of complexity, as individuals must make customization decisions about systems whose inner workings they may not fully understand or simply lack enough knowledge to understand and interpret AI systems’ behaviors and outputs (Laupichler et al., 2023; Pinski & Benlian, 2024).

2.2. Psychological ownership theory in AI systems and AI-generated outputs

Central to successful human-AI collaboration is the psychological experience of working with AI systems that can be customized to align with individual preferences and work styles. Research in human-computer interaction has established that user psychology—such as feelings of agency, control, and satisfaction—plays a crucial role in technology adoption and effective use (Gong et al., 2024; Liu, 2021). Through customization features, individuals can personalize their AI collaborators’ communication styles, areas of focus, and behavioral patterns (Gao & Liu, 2023). This ability to shape the collaborative experience may fundamentally influence individual’s psychological experience when they work and collaborate with AI systems.

A key psychological factor in this new era of human-AI collaboration is psychological ownership, which is defined as the feeling that something is “MINE” (Pierce et al., 2001, 2003). This sense of ownership is fostered through perceived control, intimate knowledge, and self-investment (Bagga et al., 2019; Pierce et al., 2003, 2004; Van Dyne & Pierce, 2004). Prior research has shown that psychological ownership can foster various positive outcomes, including increased valuation of objects and ideas (Morewedge et al., 2021; Peck & Shu, 2009), enhanced motivation and organizational commitment (Van Dyne & Pierce, 2004), improved self-esteem (Lee & Suh, 2015; Liu et al., 2012), and stronger stewardship behaviors (Peck et al., 2021). However, it can also lead to negative consequences, such as anger and interpersonal conflicts when individuals sense ownership infringements (Brown et al., 2014; Kirk et al., 2018) and resistance to change (Cocieru et al., 2019). This dual nature of psychological ownership highlights the importance of understanding its nuanced effects across various contexts.

Importantly, psychological ownership has been shown to extend beyond mere legal ownership or entitlement (Pierce et al., 2001) and can be experienced toward physical objects (Peck & Shu, 2009), abstract ideas (Baer & Brown, 2012), work projects (Cocieru et al., 2019), organizations (Brown et al., 2005; Van Dyne & Pierce, 2004), and living entities such as pets (Kirk, 2019) and humans (Yu, 2022). Pierce et al. (2001, 2003) propose that the human need for effectance, self-identity, and having a place contribute to the development of ownership feelings. These fundamental human motives help explain why individuals develop psychological ownership even in contexts where legal ownership is absent or ambiguous. As AI technology grows increasingly embedded in everyday tasks and professional workflows, psychological ownership becomes even more relevant. Recent research has shown that psychological ownership can develop toward digital goods (Atasoy & Morewedge, 2018; Watkins et al., 2016), virtual places and avatars (Lee & Chen, 2011), and even virtual assistants like Siri (Cheng, 2022).

In the context of LLMs, which often function as AI assistants, psychological ownership presents unique considerations. A critical mechanism for fostering psychological ownership is customization (Jussila et al., 2015; Kirk et al., 2015). By allowing users to modify and customize their experience with AI systems and their outputs, customization directly engages all three antecedents of psychological ownership: it grants individuals control over the AI’s output, promotes intimate knowledge of the system’s capabilities and limitations, and demands self-investment of time, effort, and creativity. Indeed, existing research in digital contexts has shown that customization can enhance user engagement and satisfaction with technology platforms (Gong et al., 2024). Through frequent human-AI interaction, such as refining prompts, modifying tones and contexts, giving feedback, and guiding outputs,

individuals may develop a deeper understanding of the AI systems' features, capacities, and limitations; invest more self through the time and effort spent customizing, and come to develop a strong sense of control over the AI system and its outputs.

In the context of human-AI collaboration, psychological ownership may influence both how individuals engage with AI systems and how they approach the collaborative work process. For example, when individuals develop strong psychological ownership of an AI system, they may invest more time in learning its capabilities, customize their inputs to guide the AI system's outputs more carefully, and take greater initiative in refining their interactions with the AI system. This enhanced engagement could lead to more effective collaboration as users treat the AI more as a customized collaborator than a generic tool. However, the role of psychological ownership in human-AI collaboration presents unique considerations. For example, if an individual develops feelings of ownership over an AI system, does this feeling of ownership extend to the outputs produced by the AI system? Additionally, how does psychological ownership influence individuals' satisfaction of the AI-generated outputs in that collaborative process, and how this may lead to responsibility attribution in human-AI collaboration?

These questions become particularly significant when considering how individuals respond to feedback or criticism of AI-assisted work. Based on psychological ownership theory and previous research on human-AI interaction, we expect customization to enhance individuals' satisfaction through two sequential psychological mechanisms: first, customization may facilitate psychological ownership of the AI system/assistant. Next, the ownership of the AI system may transfer to perceived ownership of its outputs, as users view the AI-generated content as an extension of their customized AI assistant in the collaboration process. As individuals develop psychological ownership of both the AI assistant and its outputs, this dual ownership pathway may lead them to experience stronger greater satisfaction with the outputs, potentially affecting how they attribute responsibility for successes and failures.

Given the importance how responsibility attribution may influence human-AI collaborations and potential relationships in the long-term, we next examine how customization affects responsibility attribution through psychological ownership's lens. The theory of psychological ownership offers unique insights into individual users navigate the complex boundaries between human and AI contributions in collaborative work. When outputs result from the combined efforts of human guidance and AI generation, traditional notions of accountability may become blurred. A series of important questions such as who deserves credit for successes, who bears responsibility for failures, and how users' feelings of ownership influence these responsibility attributions are raised. To better understand the effect of customization on responsibility attribution through psychological ownership's lens, we next examine how responsibility attribution operates in the context of human-AI collaboration, where the boundaries between human and AI's contributions to work outcomes can get complex and intertwined.

2.3. Attribution theory and responsibility attribution in human-AI collaboration

As AI systems become increasingly integrated into professional workflows, the question of how individuals attribute responsibility in human-AI collaborative work becomes critical (Raisch & Krakowski, 2021). Responsibility attribution is defined as the cognitive process through which individuals assign causal responsibility for outcomes, events, or states (Heider, 1958; Kelley, 1973). The process of responsibility attribution often involves determining who or what is accountable for outcomes—an issue that grows more complex when work products emerge from collaborative partnerships involving both human and AI contributions (Abedin et al., 2022; Hancock et al., 2021). A recent work by Üzümlü et al. (2025) highlights that users' perceived moral implications of AI-assisted work play an important role in understanding how they interact and adapt AI in collaborative settings, highlighting the important role of responsibility attribution in human-AI collaborations. Indeed, questions of responsibility attribution for work produced from human-AI collaborations have sparked intense debates across creative industries, academia, and professional settings (Epstein et al., 2020; Islam et al., 2024; Oravec, 2023), from AI-assisted talent recruitment processes (Islam et al., 2024) to the use of AI in peer review systems (Mollaki, 2024).

Classic attribution theory (Heider, 1958; Kelley, 1973) provides a foundation for understanding how people interpret events and assign causes. In human-computer interactions, Epstein et al. (2020)'s

research has shown that users' attributions of responsibility often depend on system transparency, perceived agency, and outcome valence. However, as AI systems are becoming increasingly sophisticated, presence of an AI collaborator introduces unique attribution challenges, as users may struggle to clearly delineate their contributions from those of the AI system (Hancock et al., 2021). As a notable example, an attorney faced professional consequences after submitting court filings containing non-existent legal cases generated by ChatGPT (Bohannon, 2023). While the attorney attributed the error to unawareness of the AI's potential for fabrication, the incident highlighted the intricate relationship between user responsibility and AI system limitations in professional contexts.

Meanwhile, insights from the literature on psychological ownership suggest that feelings of ownership may further complicate responsibility attributions in human-AI collaboration. When individuals develop psychological ownership over an object or output, they tend to feel greater responsibility for its outcomes (Pierce et al., 2001, 2003; Van Dyne & Pierce, 2004). By viewing the AI systems and their outputs as extensions of individuals' selves (Belk, 1988, 2013), these individuals may be more likely to accept criticism or negative feedback as personally relevant. When users develop strong psychological ownership of an AI assistant through customization, they may experience outputs from that AI as connected to their sense of agency and investment, leading them to treat AI-generated content as something for which they feel responsible rather than as entirely separate from their own contributions. Furthermore, it is also possible that individuals with strong psychological ownership might view AI-generated outputs more favorably, not solely based on content quality but due to their perceived personal involvement through customization. The heightened emotional attachment and involvement could impair individuals' judgment, potentially causing them to overlook flaws or biases in the AI's output. Additionally, based on fundamental attribution theory, individuals might be more likely to take credit for successful outcomes while shifting blame to the AI systems when the AI-generated outputs are challenged with negative feedback or yield undesirable results. Understanding the psychological mechanisms through which customization influences these competing effects becomes crucial for designing AI systems that promote both engagement and responsible use while mitigating potential biases in responsibility attribution.

Taken together, these perspectives imply that customization—one of the key drivers of psychological ownership—may amplify individuals' willingness to take responsibility, even in scenarios where the AI's opacity ("black box" nature) blurs agency and accountability (Lipton, 2018; Ribeiro et al., 2016). Yet, individuals might also attribute failures to the AI or external factors, especially if they perceive a lack of control over final outputs. Recognizing that responsibility often hinges on external judgments, we next consider how the source of feedback or evaluation—from human experts, non-experts, or even AI systems—can further complicate responsibility attribution in human-AI collaborations.

2.4. Source credibility theory and feedback perceptions

The source of feedback or evaluation adds another layer of complexity to responsibility attribution in human-AI collaboration. Source credibility, or the perceived credibility of the person/entity delivering the message, is shown to be a critical factor that influences message persuasiveness and interpersonal trust (Giffin, 1967; Pornpitakpan, 2004). Existing research on source credibility has shown that factors like perceived competence, authority, expertise, and trustworthiness of feedback sources can significantly influence how individuals interpret and respond to criticism (Fisher et al., 1979; Pornpitakpan, 2004). Previous research has shown that individuals are more likely to trust and incorporate feedback when it comes from perceived expert reviewers, primarily due to the heightened source credibility and domain-specific knowledge these experts are assumed to possess (DeBono & Harnish, 1988; Vermeulen & Seegers, 2009). In contrast, feedback from non-experts may be received with greater skepticism or even disregarded, particularly if recipients question the provider's authority or relevant experience (Birnbaum & Stegner, 1979; Chaiken, 1980).

However, despite the continuing integration of AI systems into daily tasks, organizational workflows, and even scientific research and practices (Binz et al., 2025), it remains unclear how people perceive "expertise" in AI. While AI review systems offer potential benefits like scalability and consistency in evaluation, they also raise questions about the credibility and authority of automated feedback

compared to human expert judgment. For example, in academic peer review, where the emergence of an increasing number of reviewers using AI to assist their reviews has sparked debate about the role of automated evaluation in quality control processes (Mollaki, 2024; Tyser et al., 2024). As the boundaries between human and AI evaluation become more blurred, how individuals respond—emotionally and behaviorally—to feedback from a human expert, a human non-expert, or an AI system is far from straightforward (Raisch & Krakowski, 2021). From an algorithmic appreciation perspective (Logg et al., 2019), individuals may view an AI reviewer as more objective and thus more credible than a human reviewer. However, individuals may also exhibit algorithm aversion and prefer human judgment over algorithmic recommendations (Jones-Jang & Park, 2023), and thus perceive an AI reviewer as less credible. More importantly, when the feedback concerns AI-generated work, individuals may view an AI reviewer as particularly qualified to evaluate other AI's outputs, or they may question the AI reviewer's ability to provide accurate feedback on a work that they have co-created with another AI assistant especially if it's their "own" AI assistant. This suggests the need to examine a novel three-way comparison: how feedback from different reviewer types—human expert, human non-expert, or AI—might affect individual users' attitudinal and behavioral reactions to such feedback.

These considerations about feedback source credibility become particularly important when examining satisfaction as an outcome in human-AI collaboration. Satisfaction with AI-generated work reflects not merely output quality but the entire collaborative experience—from the initial customization and interaction through to how users process external evaluation of their joint work. When users develop psychological ownership through customization, their satisfaction may become more resilient to criticism, yet this resilience might vary depending on whether they perceive the feedback source as credible. Understanding these dynamics is essential for designing AI systems that foster both productive collaboration and appropriate critical evaluation. Thus, our investigation examines how customization, psychological ownership, and reviewer credibility jointly shape user satisfaction in human-AI collaborative contexts.

3. Research framework and hypotheses development

Taken together, the integration of AI systems into individuals' daily lives and professional workflows can present unique challenges for understanding user psychology and behavior in human-AI collaborations. The existing literature provides a strong foundation for understanding the theories of psychological ownership (Pierce et al., 2001, 2003), attribution theory (Heider, 1958; Kelley, 1973), and source credibility theory (Pornpitakpan, 2004) in traditional contexts. However, the dynamic, collaborative nature of human-AI interaction introduces new complexities. Our research extends these traditional theories to this new domain by proposing and testing a series of hypotheses organized around the two key phases of a collaborative workflow: the initial creation process and the subsequent response to external feedback, especially when such feedback is negative.

First of all, psychological ownership theory posits that feelings of ownership emerge from control, self-investment, and intimate knowledge (Morewedge et al., 2021; Pierce et al., 2001, 2003). We argue that the act of customizing an AI assistant should directly engage these antecedents and enhance participants' psychological ownership of the AI assistant. This feeling of ownership of the AI assistant should then transfer to the work it helps create, and consequently enhances participants' satisfaction of the AI assistant's output (see Figure 1a). This leads to the following hypotheses:

H₁: AI customization will be positively associated with satisfaction with the AI output.

H_{2a}: Psychological ownership of the AI assistant will mediate the positive relationship between AI customization and satisfaction with the AI output.

H_{2b}: Psychological ownership of the AI-generated output will mediate the positive relationship between AI customization and satisfaction with the AI output.

H_{2c}: Psychological ownership of the AI assistant and psychological ownership of the AI-generated output will serially mediate the positive effect of AI customization on satisfaction.

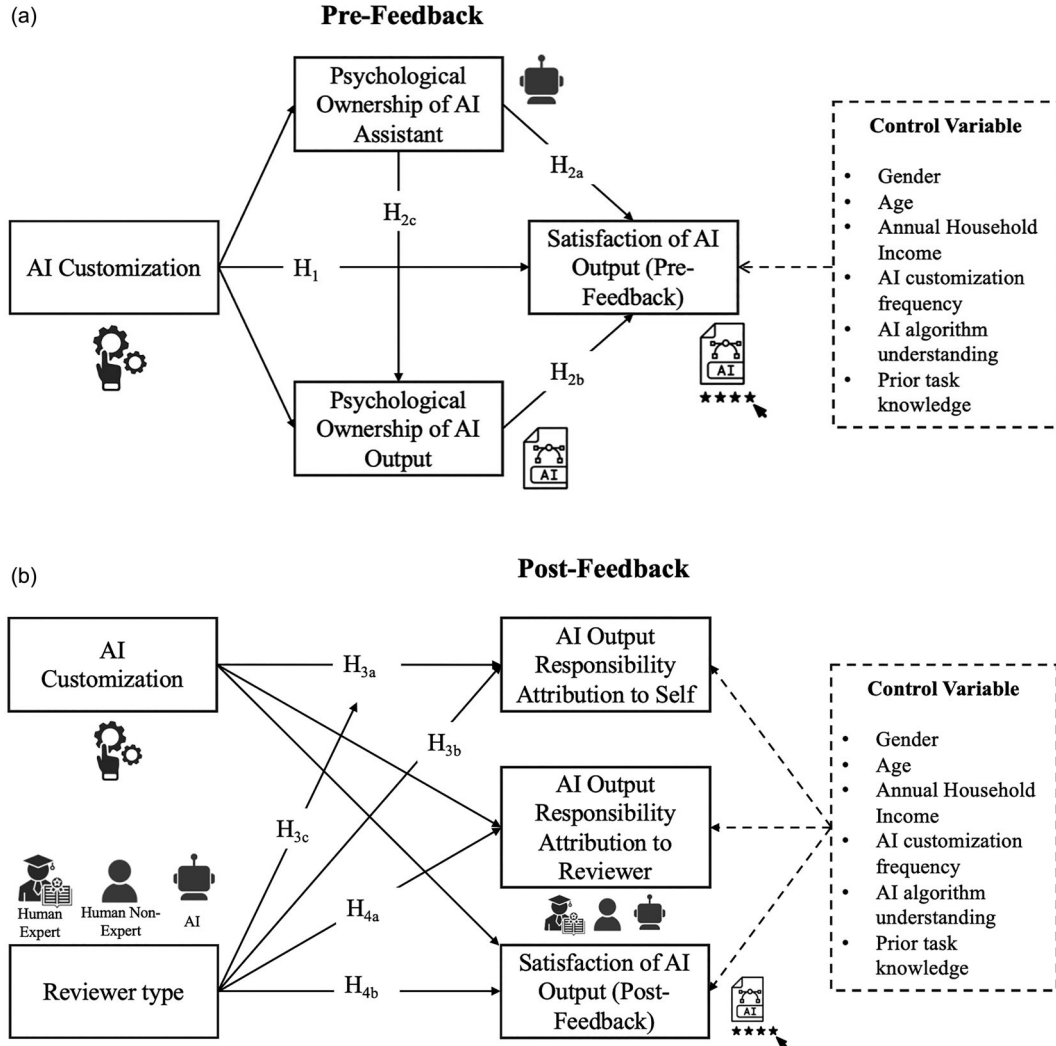


Figure 1. (a) The proposed research framework *before* participants receive negative feedback on their AI-assisted work. (b) The proposed research framework *after* participants receive negative feedback on their AI-assisted work.

Next, we argue that the introduction of external feedback, particularly negative criticism, should trigger complex psychological responses related to accountability and evaluation (Fisher et al., 1979; Pornpitakpan, 2004). Drawing on attribution theory and the principles of source credibility, we predict that both the user's sense of ownership and the reviewer's perceived expertise will influence how responsibility is attributed and how such negative feedback is received. First, we hypothesize that the heightened sense of ownership from customization will lead participants to perceive greater personal responsibility (see Figure 1(b)). Specifically:

H_{3a}: Participants who customize their AI assistant will attribute greater personal responsibility for negative outcomes compared to participants who do not.

H_{3b}: Participants who receive feedback from reviewer with high expertise will attribute greater personal responsibility for negative outcomes compared to reviewer with low expertise.

H_{3c}: There will be an interaction between AI customization and reviewer type on participants' personal responsibility attribution.

Second, we predict that the credibility of the feedback source will determine how it impacts the participants. We expect participants to be more defensive toward less credible sources and to perceive AI reviewers as holding a distinct position of expertise (see Figure 1(b)):

H_{4a}: Participants will attribute more responsibility to the reviewer for negative feedback when the reviewer has lower perceived expertise.

H_{4b}: Participants' post-review satisfaction will be less negatively affected by feedback when the reviewer has lower perceived expertise.

Finally, we propose that the initial pathway of ownership will continue to have a protective effect on satisfaction, even after criticism is received:

H₅: The positive effect of AI customization on post-feedback participant satisfaction will be serially mediated by psychological ownership of the AI assistant and psychological ownership of the AI-generated output.

By examining these theoretical gaps through our hypotheses and exploratory research questions, this study aims to advance our understanding of the psychological mechanisms underlying effective human-AI collaboration, while providing practical insights for organizations seeking to integrate AI systems in ways that enhance individual engagement and productive feedback processes. In the following sections, we delve into our methodology, empirical findings, and discussions of the theoretical and managerial implications of our research.

4. Methodology

4.1. Sample and recruitment

Seven hundred and four participants were initially recruited to the study from Prolific. To ensure a broad and generalizable sample, our recruitment criteria were intentionally inclusive, requiring only that participants be 18 years or older and fluent in English. We deliberately avoided restricting the sample to a specific age group or professional background. This approach was chosen because AI assistants are used across a wide range of demographics, and our research questions concern the fundamental psychological processes of ownership and responsibility that are not presumed to be limited to a specific population. While our participants were not required to have specific professional backgrounds, Prolific participants know they are recruited to participate in research and are shown to handle research tasks involving complex research designs consistently well, delivering higher quality data than other common crowdsourcing platforms such as Amazon's Mechanical Turk (Douglas et al., 2023; Palan & Schitter, 2018; Peer et al., 2022).

We excluded 197 respondents who failed the embedded attention checks in the study (for example, *"The chatbot I interacted with was easy to use. Please select somewhat disagree to show you are paying attention."*). The attention checks were included to ensure data quality by verifying that participants were reading instructions carefully and remained engaged throughout the experiment. This leaves a final sample size of 507 respondents ($M_{\text{age}} = 32.66$ years old; 49.7% females, 49.3% males, 1.3% other/prefer not to say). The final respondents reported that they have moderate and sufficient understanding of AI systems ($M = 4.88$, $SD = 1.25$) and customize AI systems sometimes ($M = 3.23$, $SD = 1.87$). These measures ensure that our participants demonstrate characteristics essential for meaningful engagement with our experimental task.

Our final sample size of 507 participants exceeds recommended guidelines for between-subjects experimental designs in top journals (Koschate-Fischer & Schandelmeier, 2014; Pechmann, 2019; Stoner et al., 2023). To ensure the sufficiency of our sample size, we conducted a post-hoc power analysis using G*Power 3.1 (Faul et al., 2009), aiming to detect a small-to-medium effect size ($f = 0.15$) with 0.80 power at $\alpha = 0.05$. The results indicated that a minimum sample of 428 participants was required to achieve adequate statistical power. As our final sample size consisted of 507 participants, it was sufficiently powered to detect the hypothesized effects.

4.2. Research design and experimental procedure

This experiment employed a 2 (AI Customization: Yes vs. No) $\times$ 3 (Reviewer Type: Human Novice vs. Human Expert vs. AI) between-subjects design. Participants were recruited through Prolific and

randomly assigned to one of six experimental conditions. All participants were asked to imagine they were research associates in a biotechnology firm tasked with creating an mRNA vaccine technology overview report using ChatGPT 4o mini. We selected this specific AI model because it was freely accessible without requiring account registration and was widely used at the time of the study, ensuring consistent access for all participants.

For the AI customization manipulation, participants in the *customization* condition were instructed to customize their AI assistant before generating the report by giving it a nickname and selecting preferences such as tone and areas of focus. In contrast, participants in the *no customization* condition were provided with a standardized prompt (“Please help me write an overview report on mRNA vaccine technology that outlines key concepts, current applications, recent advancements, and potential future implications.”) and were instructed to not customize the AI so their interaction with a standard, non-customized AI can be studied (see Table 1 for more details). After generating and submitting their reports, all participants completed initial measures assessing their attitudes and perceptions about the AI and its output.

Next, participants were presented with a scenario where the mRNA vaccine technology report they just generated was reviewed and received negative feedback. The source of this feedback varied by condition: participants were told the review came from either a *non-expert* colleague with no biotechnology experience, a highly respected *expert* with extensive mRNA research experience, or an AI-powered internal review system. The negative feedback was consistent across conditions, suggesting the report lacked necessary detail and fell short of professional scientific communication standards (see Table 1 for more details). Participants completed several measures both before and after receiving the negative feedback.

4.3. Measures

Participants completed several measures both before and after receiving the negative feedback. To ensure the quality of our measurements, all multi-item metrics’ reliability was assessed with Cronbach’s alpha, with all scales demonstrating excellent internal consistency. The key constructs were established by adapting scales from previously validated research. Prior to receiving the negative feedback from the reviewer, we asked participants to report their psychological ownership (assessed using two 4-item scales adapted from Van Dyne & Pierce, 2004) toward both the AI system (e.g., “The version of the chatbot that assisted me with the report is MY chatbot.”; Cronbach’s $\alpha = 0.89$) and the report it helped generated (e.g., “The report is MY work.”; Cronbach’s $\alpha = 0.88$). We also measured participants’ satisfaction with the AI’s output using a 3-item scale (e.g., “I am satisfied with the report I created in this task.”; Cronbach’s $\alpha = .87$). The 3-item satisfaction scale, adapted from well-established satisfaction measures (Churchill & Surprenant, 1982; Parasuraman et al., 1991), was specifically designed for the current study to directly assess participants’ satisfaction with the report generated in the experimental task. Participants also rated their real-life knowledge or expertise on mRNA vaccine technology (1 item).

Following the negative feedback, participants were asked to distribute responsibility for the negative feedback among themselves, the AI assistant, and the reviewer, with responsibility attribution scores totaling 100%. We also measured participants’ satisfaction with the AI’s output again using the same 3-item scale (Cronbach’s $\alpha = 0.95$), which allows us to capture the change of satisfaction perception pre- and post-feedback.

Additionally, as manipulation checks, we measured perceived AI customization (2 items, $r = 0.81$) and perceived expertise of the reviewer (1 item). Other control variables included participants’ frequency of AI customization (1 item), self-reported knowledge of mRNA vaccines (1 item), and understanding of AI algorithms (4 items, Cronbach’s $\alpha = 0.89$). The scale items for all the measures included in the study can be found in Table 2. Lastly, we asked participants to report their gender, age, and annual household income level so we could control for these basic demographic factors in our analyses.

4.4. Statistical techniques used

The data were analyzed using SPSS Version 28. To test our hypotheses, we employed several statistical techniques. We conducted a series of one-way analyses of covariance (ANCOVAs) to examine the

Table 1. Experimental vignettes for AI customization and reviewer type manipulations.

AI customization (yes, No)
[Standard Instructions] Imagine you are a research associate working for a biotechnology firm. Your company has tasked you to draft an overview report on mRNA vaccine technology. An overview report is like a roadmap that explains key concepts, current applications, recent advancements, and potential future implications of the specified technology. The report should outline these main points to educate new employees and team members who may not have specialized knowledge on mRNA vaccine technology and its recent advancements and relevance to the company. You will be using ChatGPT 4o mini to complete this task following the steps below: Step 1: Open a new web page and visit ChatGPT 4o mini's interface here: https://chatgpt.com (please manually paste this link to a new web page) This link will take you directly to the chat interface. Please DO NOT close this survey and leave this page open when you interact with ChatGPT 4o mini. *You can access ChatGPT 4o mini without registering an account. If you already have access to other ChatGPT models (e.g., GPT-4o/o1/o1 pro), we kindly ask you to sign out of your account and open a new conversation session for the purpose of this task. This ensures your session is independent and that you are interacting with the model in a fresh state. Please also ensure you are interacting with GPT-4o mini for the purpose of this task. [Customization: Yes] Step 2: Once on the ChatGPT interface, you can start your interaction based on the task provided above. Before asking ChatGPT to draft the onboarding document, please take some time to customize your ChatGPT by giving your ChatGPT a nickname (e.g., 'I would like to call you Alex.') and selecting preferences for the AI, such as tone, and areas of focus (e.g., 'You will role play as a research associate who is working for a biotechnology firm and is specialized in creating overview reports on mRNA vaccine technology.'). These are just some examples, and please use your OWN words and create your own instructions when customizing your own ChatGPT. You can then engage in further dialogues with your customized AI to tweak and finalize the report. [Customization: No] Step 2: Once on the ChatGPT interface, you can start your interaction based on the task provided above. You should copy and paste the following prompt into the dialogue window as the first thing when you engage ChatGPT: 'Please help me write an overview report on mRNA vaccine technology that outlines key concepts, current applications, recent advancements, and potential future implications.' You can then engage in further dialogues to tweak and finalize the report. However, please do NOT customize the AI (e.g., telling the AI to use a specific tone, or telling the AI to roleplay/adopt an area of expertise). This is because we are interested in studying your interaction with a standard, non-customized AI.
Reviewer Type (Human Expert, Human Non-Expert, AI)
[Human Expert Reviewer] Now, imagine that you have submitted the mRNA vaccine technology report to your company. A highly respected expert in biotechnology with extensive experience in mRNA research reviews the report and expresses significant concerns. They suggest the report lacks necessary detail in several key areas and includes explanations that fall short of the clarity expected in professional scientific communications. In short, they give the report a negative evaluation. [Human Non-Expert Reviewer] Now, imagine that you have submitted the mRNA vaccine technology report to your company. A colleague who is NOT specialized in biotechnology and has NO experience in mRNA research reviews the report and expresses significant concerns. They suggest the report lacks necessary detail in several key areas and includes explanations that fall short of the clarity expected in professional scientific communications. In short, they give the report a negative evaluation. [AI Reviewer] Now, imagine that you have submitted the mRNA vaccine technology report to your company. Your company has an internal review system powered by AI to evaluate technical documents. The AI system reviews the report and expresses significant concerns. The AI system suggests the report lacks necessary detail in several key areas and includes explanations that fall short of the clarity expected in professional scientific communications. In short, it gives the report a negative evaluation.

effects of AI customization on participants' psychological ownership of the AI assistant and the outputs. Then, a two-way multivariate analysis of covariance (MANCOVA) was used to examine the effects of AI customization and reviewer type on the three interdependent responsibility attribution measures. We also used a series of two-way ANCOVAs to test the independent effects of our AI customization and reviewer type on key dependent variables (psychological ownership and user satisfaction). Next, a mixed-design ANCOVA was conducted to assess the change in satisfaction over time (pre- vs. post-feedback). Finally, mediation and serial mediation pathways were tested using the PROCESS macro for SPSS (Model 4 and Model 6) with 10,000 bootstrap samples, as described by Hayes (2017).

5. Results

5.1. Preliminary analyses and manipulation checks

First, we examined whether our mRNA vaccine technology task successfully created a context where participants had limited prior knowledge or experience. This is because of two important considerations: first, we wanted to ensure that participants do not possess enough knowledge or experience on this topic, and would rely more on the AI's capabilities rather than their own expertise; second, users' attitudes and behavioral patterns may differ between expert and non-experts especially when it comes to using AI to assist their work (Laupichler et al., 2023; Pinski & Benlian, 2024), and high domain expertise might further confound the effects of AI customization and reviewer feedback on

Table 2. Dependent measures and items.

Psychological Ownership of the AI Assistant (phrased as a “chatbot” in the measurement items) (adapted from Van Dyne & Pierce, 2004). Cronbach's $\alpha = 0.89$.

1 = Strongly Disagree; 7 = Strongly Agree

1. The version of the chatbot that assisted me with the onboarding document is MY chatbot.
2. I sense that the chatbot is MY chatbot.
3. I feel a very high degree of ownership for the chatbot that assisted me.
4. It is hard for me to think about the chatbot that assisted me as MINE.* (Reverse-coded)

Psychological Ownership of the AI-Generated Report (adapted from Van Dyne & Pierce, 2004). Cronbach's $\alpha = 0.88$.

1 = Strongly Disagree; 7 = Strongly Agree

1. The report is MY work.
2. I sense that the report is MY work.
3. I feel a very high degree of ownership for the report.
4. It is hard for me to think about the report as MINE.* (Reverse-coded)

Satisfaction of the AI-Generated Report (pre-feedback: Cronbach's $\alpha = .87$; post-feedback: Cronbach's $\alpha = 0.95$; adapted from Churchill & Surprenant, 1982; Parasuraman et al., 1991)

1 = Strongly Disagree; 7 = Strongly Agree

1. I am satisfied with the report I created in this task.
2. The content of the report meets my expectations.
3. I like the report I created in this task.

Responsibility Attribution

Please distribute the total responsibility for the negative feedback received on the mRNA vaccine technology report. The total must add up to 100%.

- Yourself (as the research associate): _____
- The Chatbot/ChatGPT AI tool you used: _____
- The Reviewer who provided that negative feedback: _____

Perceived Expertise of the Reviewer, 1 = Not at all expert; 7 = Extremely expert

1. How would you rate the expertise of the reviewer who reviewed the report?

Self-Reported Knowledge/Expertise of mRNA vaccines, 1 = Extremely low; 7 = Extremely high

1. Outside of this scenario, how would you rate your real-life knowledge or expertise on mRNA vaccine technology?

Self-Reported Frequency of AI Customization in Daily Use, 1 = Never; 7 = Always

1. How often do you customize (e.g., naming the AI, specifying the AI's knowledge content, skills, tone of communication, etc.) a Chatbot/generative AI agent to fit your need?

Self-Reported Knowledge/Understanding of AI Algorithms, Cronbach's $\alpha = 0.89$; 1 = Strongly Disagree; 7 = Strongly Agree

1. I understand how this AI processes information.
2. I can explain how this AI generates outputs.
3. I understand the limitations of this AI's capabilities.
4. I understand the basic principles driving this AI's behaviors.

Perceived AI Customization Level (Study Task-Specific) $r = 0.81$; 1 = Strongly Disagree; 7 = Strongly Agree

1. I customized the chatbot before using it for the task.
2. The chatbot I interacted with was customized based on my instructions.

Note: all measures were taken on a 7-point Likert scale, except for responsibility attribution.

psychological ownership and responsibility attribution. A pretest with undergraduate students ($n = 48$, $M_{\text{age}} = 22.44$ years old, 87.5% females) from a university in North America confirmed that this topic was sufficiently unfamiliar ($M = 2.79$, $SD = 1.58$, on a 7-point scale). This was further validated in our main study, where participants reported low levels of expertise in mRNA vaccine technology ($M = 2.95$, $SD = 1.56$, on a 7-point scale).

We also examined the effectiveness of our AI customization manipulation. The two items measuring perceived AI customization were highly correlated ($r = 0.81$, $p < 0.001$) and were averaged to create a composite score. A 2 (AI Customization) $\times$ 3 (Reviewer Type) ANOVA on this measure revealed a significant main effect of the AI customization manipulation ($F(1, 501) = 734.82$, $p < 0.001$, $\text{partial } \eta^2 = 0.595$). As intended, participants in the customization condition reported higher levels of perceived AI customization ($M_{\text{customization}} = 5.97$, $SD = 1.06$) compared to those in the no-customization condition ($M_{\text{no-customization}} = 2.49$, $SD = 1.71$). Neither the main effect of reviewer type ($F(2, 501) = 0.17$, $p = 0.85$, $\text{partial } \eta^2 = 0.001$) nor the interaction effect ($F(2, 501) = 1.62$, $p = 0.20$, $\text{partial } \eta^2 = 0.006$) was significant, indicating that our customization manipulation worked as intended independently of the reviewer manipulation.

Additionally, we examined the effectiveness of our reviewer type manipulation through a 2 (AI Customization) $\times$ 3 (Reviewer Type) ANOVA on the perceived expertise of the reviewer. Results revealed a significant main effect of reviewer type, $F(2, 501) = 114.51$, $p < 0.001$, $\text{partial } \eta^2 = 0.314$. Neither the main effect of AI customization ($F(1, 501) = 0.16$, $p = 0.69$, $\text{partial } \eta^2 < 0.001$) nor the interaction effect ($F(2, 501) = 0.23$, $p = 0.80$, $\text{partial } \eta^2 = 0.001$) was significant. Post-hoc comparisons using Tukey's HSD test indicated that all three conditions differed significantly from each other (all p -values < 0.001). As intended, the human expert reviewer was perceived as having the highest expertise

($M_{human\ expert} = 6.13$, $SD = 1.96$), followed by the AI reviewer ($M_{AI} = 5.04$, $SD = 1.99$), and the human non-expert reviewer was perceived as having the lowest expertise ($M_{human\ non-expert} = 2.97$, $SD = 1.94$).

Due to the complexity of this experiment and its data structure, in the following sections concerning the core analyses, we present and discuss the findings in two sections: pre-feedback and post-feedback results. This allows us to better illustrate how AI customization and reviewer type influenced psychological ownership and responsibility attribution over time.

5.2. Pre-feedback results

5.2.1. Psychological ownership of the AI assistant

We first examined the effect of AI customization on psychological ownership toward the AI assistant, controlling for participants' age, gender, income, general tendency to customize AI, self-reported knowledge of mRNA vaccines, and their self-rated understanding of AI algorithms. The one-way ANCOVA revealed that our AI customization manipulation significantly increased psychological ownership toward the AI ($F(1, 499) = 22.60$, $p < 0.001$, $partial\ \eta^2 = 0.043$).¹ Participants in the customization condition reported higher levels of psychological ownership toward the AI assistant ($M_{customization} = 3.48$, $SD = 1.62$) compared to those in the no-customization condition ($M_{no-customization} = 2.95$, $SD = 1.47$).

5.2.2. Psychological ownership of the AI-generated report

Next, we examined the effect of AI customization on participants' psychological ownership toward the AI-generated mRNA vaccine technology report. Results from the one-way ANCOVA test indicated that AI customization also had a significant positive effect on psychological ownership toward the generated report ($F(1, 499) = 7.27$, $p = 0.01$, $partial\ \eta^2 = 0.014$). Participants who customized the AI assistant reported higher levels of psychological ownership toward the output report ($M_{customization} = 3.04$, $SD = 1.46$) compared to those who did not customize the AI assistant ($M_{no\ customization} = 2.78$, $SD = 1.42$). This effect remained significant after controlling for demographic variables and individual differences in AI experience and task knowledge.²

To test whether psychological ownership of the AI assistant mediated the effect of customization on report ownership, we conducted a mediation analysis using PROCESS (Model 4; Hayes, 2017) with 10,000 bootstrap samples, controlling for all demographic variables and individual differences. The analysis revealed a significant indirect effect of AI customization on report ownership through psychological ownership of the AI assistant (indirect effect = 0.34, $SE = 0.076$, 95% CI [0.1962, 0.4980]). After accounting for this mediation, the direct effect of customization on report ownership became non-significant (direct effect = -0.025, $SE = 0.096$, $p = 0.79$), suggesting full mediation. This suggests that customization increased psychological ownership of the AI assistant, which in turn increased psychological ownership of the AI-generated report.

5.2.3. Satisfaction with the AI-generated report (pre-feedback)

Finally, we tested participants' satisfaction with the AI-generated report using the same one-way ANCOVA test as consistent with the previous analyses. Our results revealed that AI customization had a significant positive effect on satisfaction with the generated report, $F(1, 499) = 4.32$, $p = 0.04$, $partial\ \eta^2 = 0.009$. Participants who customized the AI reported higher satisfaction with the report ($M_{customization} = 6.04$, $SD = 0.74$) compared to those who did not customize the AI ($M_{no-customization} = 5.92$, $SD = 0.86$). Thus, H_1 was supported. This effect remained significant after controlling for demographic variables and individual differences in AI experience and task knowledge.³

We first test the mediation effects of customization on satisfaction through participants' psychological ownership of the AI assistant and the AI generated report separately. Using Hayes' PROCESS (Model 4) with 10,000 bootstrap samples and the same control variables, we found participants' psychological ownership of the AI assistant significantly mediated the effect of customization on satisfaction (indirect effect = 0.07, $SE = 0.020$, 95% CI [0.0307, 0.1097]). Thus, H_{2a} was supported. We also found participants' psychological ownership of the AI-generated report significantly mediated the effect of customization on satisfaction (indirect effect = 0.05, $SE = 0.021$, 95% CI [0.0113, 0.0926]). Therefore, H_{2b} was supported.

Lastly, we conducted a serial mediation analysis using Hayes' PROCESS (Model 6) with 10,000 bootstrap samples, controlling for the same variables as stated previously. The analysis revealed a significant sequential indirect effect of customization on satisfaction through participants' psychological ownership of the AI assistant and the AI-generated report (indirect effect = 0.04, SE = 0.016, 95% CI [0.0179, 0.0805]), supporting H_{2c} . Specifically, within the model, PO of the AI assistant was a strong predictor of PO of the AI-generated report ($\beta = 0.58$, $p < 0.001$). Neither the direct effect of customization on satisfaction (direct effect = 0.08, $p = 0.23$) nor the other indirect paths (PO of the AI only: indirect effect = 0.02, 95% CI [-0.0144, 0.0589]; PO of the report only: indirect effect = -0.003, 95% CI [-0.0304, 0.0221]) were significant, suggesting that the sequential mediation path fully explains the relationship between customization and satisfaction.

5.2.4. Discussions of the pre-feedback results

These pre-feedback findings revealed important understanding of the psychological mechanisms through which AI customization influences individual users' psychological ownership and satisfaction in human-AI collaboration. The sequential mediation results suggested a rather precise psychological pathway: customization first builds ownership of the AI system itself, which then transfers to ownership of its outputs, ultimately leading to higher satisfaction. This chain of effects suggests that the benefits of AI customization aren't direct or simple but operate through a specific sequence of psychological states. This finding demonstrates that in human-AI collaboration, ownership can transfer across different targets (from tool to output) in a structured way. Moreover, it suggests that for individuals to feel satisfied with AI-generated outputs, they must first develop ownership feelings toward both the AI system and its outputs—neither form of ownership alone is sufficient.

Building on these insights on how AI customization fosters user ownership and satisfaction in AI-assisted tasks like the one presented in the study, we now turn to the post-feedback analyses. In the next section, we examine how reviewer input further shapes these outcomes once participants receive external evaluation and guidance on their AI-generated work.

5.3. Post-Feedback results

5.3.1. Responsibility attribution

We conducted a two-way MANCOVA to examine the effects of AI customization and reviewer type on how participants attributed responsibility for the negative feedback among themselves, the AI, and the reviewer (responsibilities were divided between the three parties and totaling 100), controlling for demographic variables and individual differences. The multivariate tests revealed significant main effects of both AI customization (Wilks' $\lambda = 0.988$, $F(2, 494) = 3.08$, $p = 0.047$, *partial* $\eta^2 = 0.012$) and reviewer type (Wilks' $\lambda = 0.951$, $F(4, 988) = 6.26$, $p < 0.001$, *partial* $\eta^2 = 0.025$). The interaction between customization and reviewer type was not significant (Wilks' $\lambda = 0.995$, $F(4, 988) = 0.67$, $p = 0.612$, *partial* $\eta^2 = 0.003$).⁴ Follow-up univariate analyses revealed several interesting responsibility attribution results when participants received negative feedback on the report (see Figure 2a, 2b, 2c).

5.3.1.1. Responsibility attributed to self. For participants' perceived responsibility of self, AI customization had a significant effect ($F(1, 495) = 6.124$, $p = 0.01$, *partial* $\eta^2 = 0.012$), with participants in the customization condition taking more responsibility to themselves ($M_{\text{customization}} = 52.03$, $SD = 30.50$) compared to those in the no-customization condition ($M_{\text{no-customization}} = 44.57$, $SD = 30.95$), supporting H_{3a} . The interaction between reviewer type and customization ($F(2, 495) = 0.93$, $p = 0.40$, *partial* $\eta^2 = 0.004$) and the main effect of reviewer type ($F(1, 495) = 1.42$, $p = 0.24$, *partial* $\eta^2 = 0.003$) were non-significant, suggesting that customization was the only factor that influenced participants' responsibility attributed to themselves. Therefore, H_{3b} and H_{3c} are not supported.

5.3.1.2. Responsibility attributed to the reviewer. There was also a significant effect of reviewer type ($F(2, 495) = 11.02$, $p < 0.001$, *partial* $\eta^2 = 0.043$). Post-hoc tests with Tukey's HSD suggests that, when attributing reviewer responsibility, participants assigned significantly more responsibility or blame to

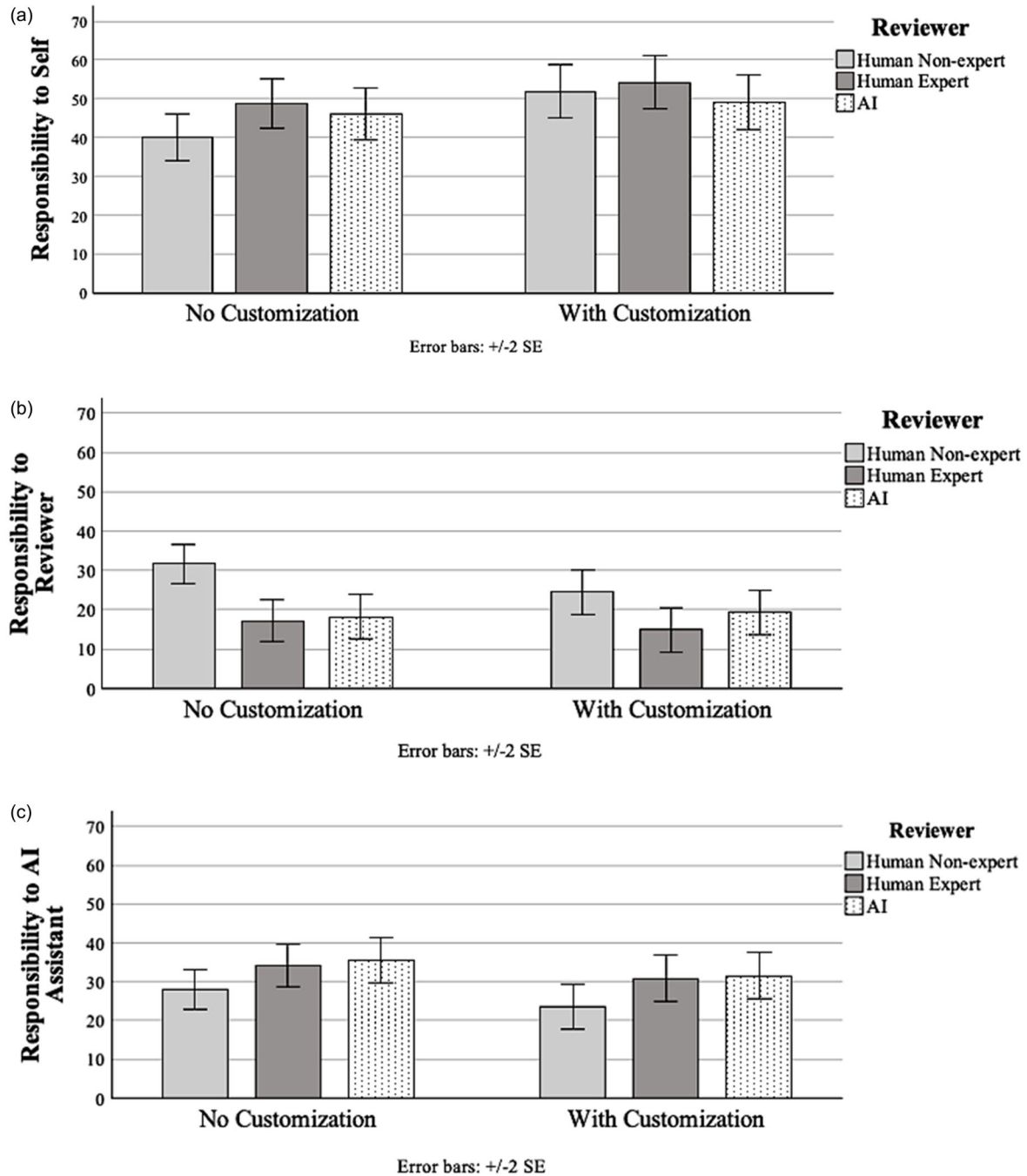


Figure 2. (a) The effect of customization and reviewer identity on responsibility attributed to participants themselves. (b) The effect of customization and reviewer identity on responsibility attributed to the reviewer who provided the negative feedback. (c) The effect of customization and reviewer identity on responsibility attributed to the AI assistant (ChatGPT 4o-mini).

the human non-expert reviewer ($M_{human \text{ non-expert}} = 28.83$, $SD = 29.84$) compared to both the human expert reviewer ($M_{human \text{ expert}} = 16.22$, $SD = 21.60$, $p < 0.001$) and the AI reviewer ($M_{AI} = 18.34$, $SD = 23.12$, $p < 0.001$), supporting H_{4a} . Notably, responsibility attributed to AI reviewer did not differ significantly from that attributed to human expert reviewer ($p = 0.73$), suggesting that participants viewed AI and expert human reviewers as similarly credible sources of feedback when evaluating an AI-generated report. The interaction between reviewer type and customization ($F(2, 495) = 1.15$, $p = 0.32$, $\text{partial } \eta^2 = 0.005$) and the main effect of customization ($F(1, 495) = 1.52$, $p = 0.22$, $\text{partial } \eta^2 = 0.003$) were nonsignificant.

5.3.1.3. Responsibility attributed to the AI assistant. For participants' perceived responsibility of the AI that assisted them with generating the report, there was a significant effect of reviewer type ($F(2, 495) = 4.482, p = 0.01, \text{partial } \eta^2 = 0.018$). Post-hoc tests with Tukey's HSD revealed distinct patterns in how participants attributed responsibility after receiving negative feedback on their mRNA vaccine reports. Participants whose reports were reviewed by an AI system attributed significantly more responsibility to the AI that assisted them to complete the report ($M_{AI} = 33.65, SD = 26.19$) compared to those whose reports were reviewed by a human non-expert ($M_{human \text{ non-expert}} = 26.23, SD = 24.51, p = 0.03$). Additionally, AI responsibility attribution in the AI reviewer condition did not differ significantly from the human expert reviewer condition ($M_{human \text{ expert}} = 32.40, SD = 27.53, p = 0.90$). These findings suggest that participants might hold the AI assistant less accountable for negative feedback when the feedback came from a non-expert reviewer. However, there was no significant difference between the human expert reviewer condition and the human non-expert reviewer condition ($p = 0.07$). The interaction between reviewer type and customization ($F(2, 495) = 0.03, p = 0.97, \text{partial } \eta^2 < 0.001$) and the main effect of customization ($F(1, 495) = 2.87, p = 0.09, \text{partial } \eta^2 = 0.006$) were nonsignificant.

While the next reasonable step is to examine potential mediating mechanisms in responsibility attribution, the responsibility attribution variables are compositional and interdependent (i.e., participants were asked to distribute a total of 100% responsibility among themselves, the AI assisted them with the report, and the reviewer). This violates the fundamental assumption of independence in traditional mediation analysis, potentially leading to biased estimates of indirect effects and require more specialized analytic approaches for experimental data (Aitchison, 1982; Aung et al., 2020; Hayes, 2017). Due to these limitations, we shift gears to focus on the next key variable in our measure: participants' satisfaction with the AI-generated report after it received criticism.

5.3.2. Satisfaction with the AI-generated report (post-feedback)

A two-way ANCOVA on post-feedback satisfaction revealed significant main effects of both AI customization ($F(1, 495) = 7.41, p = 0.01, \text{partial } \eta^2 = 0.015$) and reviewer type ($F(2, 495) = 9.61, p < 0.001, \text{partial } \eta^2 = 0.037$), controlling for the same variables as the previous tests.⁵ The interaction between customization and reviewer type was not significant ($F(2, 495) = 0.44, p = 0.65, \text{partial } \eta^2 = 0.002$). After receiving negative feedback on their mRNA vaccine report, satisfaction levels varied significantly by both customization condition and reviewer type. Participants who had customized their AI assistant maintained higher satisfaction with their report ($M_{customization} = 4.88, SD = 1.67$) compared to those who did not customize the AI ($M_{no-customization} = 4.55, SD = 1.63$). Post-hoc analysis showed that participants maintained significantly higher satisfaction when their report was reviewed by a human non-expert ($M_{human \text{ non-expert}} = 5.09, SD = 1.49$) compared to both a human expert ($M_{human \text{ expert}} = 4.36, SD = 1.80, p < 0.001$) and an AI reviewer ($M_{AI} = 4.63, SD = 1.59, p = 0.03$), supporting H_{4b} . Interestingly, satisfaction levels did not differ significantly between human expert and AI reviewers ($p = 0.29$), suggesting that participants responded similarly to criticism from these two sources. These results indicate that the source of negative feedback significantly influences how participants maintain satisfaction with their work, and perceived expertise directly impacts participants' considerations given to such criticism.

Furthermore, a serial mediation analysis (PROCESS Model 6; Hayes, 2017) with 10,000 bootstrap samples with the same control variables. The analysis revealed a significant sequential indirect effect of customization on post-feedback satisfaction through psychological ownership of the AI and the AI-generated report in series (indirect effect = 0.07, SE = 0.03, 95% CI [0.0198, 0.1308]). Thus, H_5 was supported. Customization also indirectly influenced post-feedback satisfaction through psychological ownership of the AI alone (indirect effect = 0.11, SE = 0.05, 95% CI [0.0317, 0.2133]). Neither the direct effect of customization on satisfaction (direct effect = 0.19, $p = 0.18$) nor the indirect effect through document ownership alone (indirect effect = -0.005, 95% CI [-0.0469, 0.0402]) was significant. Comparing the serial mediation analysis with participants' pre-feedback satisfaction scores, whereas the path through psychological ownership of the AI was not significant, this might suggest that

psychological ownership plays an even more important and salient role for maintaining satisfaction after people receive negative criticism.

While reviewer type significantly affected post-feedback satisfaction, additional analyses revealed that this effect was not mediated through psychological ownership mechanisms (all indirect effects' 95% CIs included zero). This suggests that the psychological processes underlying responses to feedback from different sources (human experts, novices, or AI) operate through different mechanisms than those involved in AI customization. Specifically, while customization builds satisfaction through developing psychological ownership, the impact of reviewer expertise on satisfaction likely operates through other mechanisms such as perceived credibility or legitimacy of the feedback source.

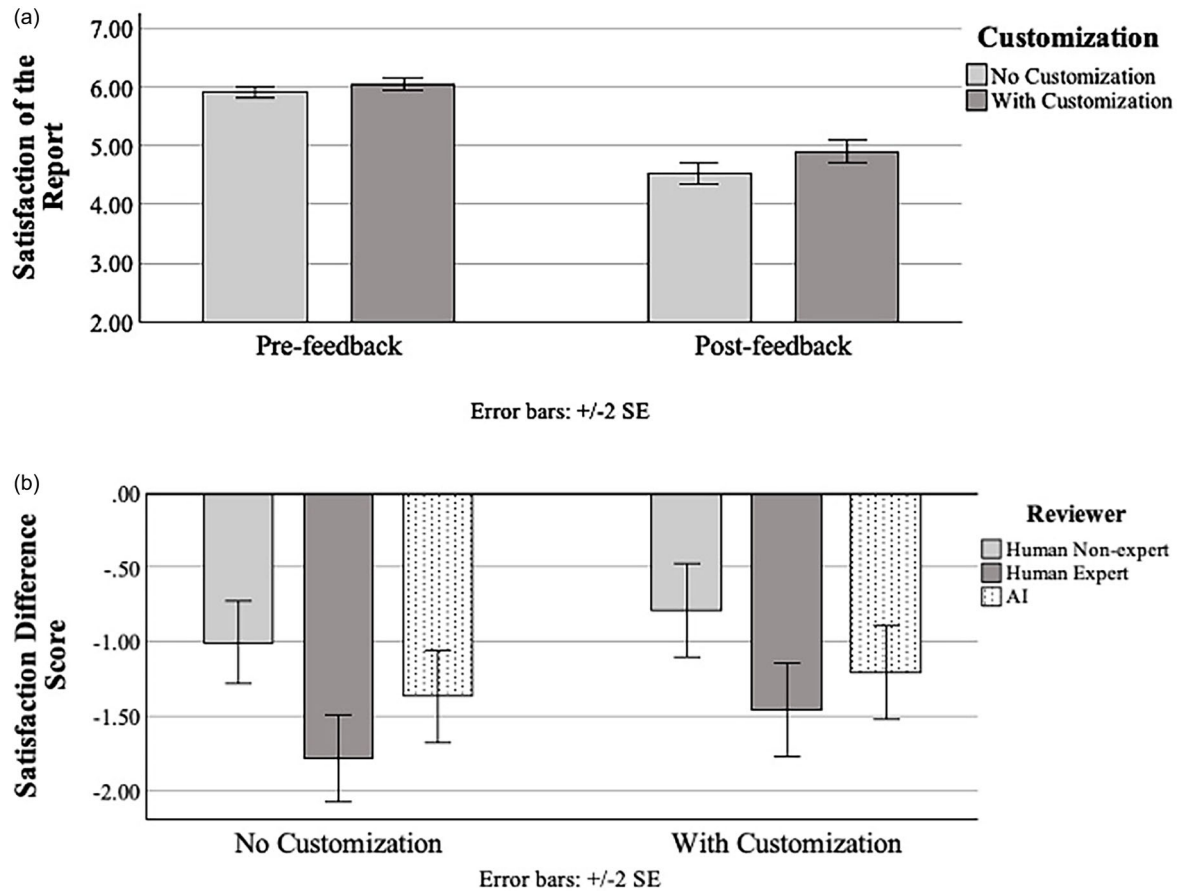
To follow up on this observation, we conducted a mediation analysis (PROCESS Model 4; Hayes, 2017) with 10,000 bootstrap samples to test whether the effect of reviewer type on satisfaction would be mediated through participants' perceived expertise of the reviewer. Indeed, the mediation results showed that the effect of reviewer type was fully mediated by perceived expertise of the reviewer (indirect effects: human expert vs. non-expert = -0.78 , 95% CI [-1.0315 , -0.5433]; AI vs. non-expert = -0.52 , 95% CI [-0.7043 , -0.3458]), with direct effects becoming non-significant. Taken together, the mediation results seem to suggest that while customization builds satisfaction through developing psychological ownership, the impact of reviewer type operates through perceived expertise and credibility of the feedback source.

5.3.3. Change of satisfaction with the AI-generated report

First, we conducted a 2 (AI Customization) $\times$ 3 (Reviewer Type) $\times$ 2 (Time: Pre-Feedback, Post-Feedback) mixed-design ANCOVA, with Time as the within-subjects factor to see if the change in participants' satisfaction with the AI-Generated Report differs. The same control variables were included as covariates.⁶ The findings revealed a significant main effect of time as the within-subjects factor ($F(1, 495) = 14.05$, $p < 0.001$, $partial \eta^2 = 0.028$), with overall satisfaction decreasing from pre-feedback ($M_{pre-feedback} = 5.97$, $SD = 0.81$) to post-feedback ($M_{post-feedback} = 4.71$, $SD = 1.66$). The interaction between time and AI customization was also significant ($F(1, 495) = 3.85$, $p = 0.05$, $partial \eta^2 = 0.008$). Participants who had customized their AI assistant showed a smaller decline in satisfaction (Pre-feedback: $M_{customization} = 6.04$, $SD = 0.74$; Post-feedback: $M_{customization} = 4.88$, $SD = 1.67$) compared to those who had not customized the AI (Pre-feedback: $M_{no-customization} = 5.92$, $SD = 0.86$; Post-feedback: $M_{no-customization} = 4.55$, $SD = 1.63$). The interaction between time and reviewer type was nonsignificant ($F(1, 495) = 0.23$, $p = 0.80$, $partial \eta^2 = 0.001$). The three-way interaction between time, AI customization, and reviewer type was also nonsignificant ($F(1, 495) = 0.12$, $p = 0.89$, $partial \eta^2 = 0.000$). These findings suggest that the overall patterns of change in satisfaction from pre-feedback to post-feedback differed for those who customized (vs. did not customize) the AI assistant, but did not differ across the three reviewer type conditions (see Figure 3(a)).

The results from the mixed ANCOVA suggested that participants' satisfaction of their mRNA report decreased as a result of receiving negative feedback on the report. Those who did not customize the AI assistant experienced a larger decrease in their satisfaction than those who did. However, while this repeated-measures analysis reveals the general trajectory of satisfaction over time, it does not fully capture whether the *magnitude* of this change differed significantly as a function of both customization and reviewer type. Additionally, as the Reviewer Type manipulation was introduced only after the pre-feedback satisfaction measurement, this means reviewer type could not have interacted with the satisfaction scores over time, and the non-significant interaction was expected.

To appropriately assess the whether the magnitude of change in satisfaction differed across both customization and reviewer type conditions, we calculated a satisfaction difference score between participants' pre-feedback and post-feedback satisfaction scores. This approach follows research highlighting the utility of difference scores (Rogosa & Willett, 1983; Zimmerman & Williams, 1982) and allows us to acknowledge the complexities inherent in analyzing pretest-posttest data solely through repeated-measures (Huck & McLean, 1975). Specifically, we computed this satisfaction difference score by subtracting pre-feedback from post-feedback satisfaction scores (negative values indicating declines in satisfaction; see Figure 3(b)).



Note: Larger negative values indicate greater declines in satisfaction.

Figure 3. (a) Changes in satisfaction of the report before and after participants received negative feedback on the report between customization and no-customization conditions. (b) The effects of customization and reviewer type on satisfaction difference scores (post-feedback—pre-feedback).

A 2 (AI Customization) $\times$ 3 (Reviewer Type) ANCOVA with the same control variables⁷ revealed a significant main effect of reviewer type ($F(2, 495) = 11.63, p < 0.001, \text{partial } \eta^2 = 0.045$). Participants showed the largest decline in satisfaction when receiving feedback from a human expert reviewer ($M_{\text{human expert}} = -1.61, SD = 1.57$), which differed significantly from both a human non-expert reviewer ($M_{\text{human non-expert}} = -0.90, SD = 1.27, p < 0.001$) and an AI reviewer ($M_{\text{AI}} = -1.32, SD = 1.38, p = 0.02$). While participants in the AI reviewer condition reported a smaller decline in satisfaction compared to participants in the human expert condition, this difference was not statistically significant ($p = 0.15$). Interestingly, although the previous mixed ANCOVA test suggested that participants who customized the AI showed a pattern of smaller decline in their satisfaction levels across-time, the main effect of customization on the specific magnitude of scores did not reach statistical significance ($F(1, 495) = 3.50, p = 0.06, \text{partial } \eta^2 = 0.007$), suggesting that AI customization might provide a certain degree of buffering against the negative effects of feedback, but the magnitude of this buffering remains limited. The interaction between customization and reviewer type was not significant ($F(2, 495) = 0.14, p = 0.868, \text{partial } \eta^2 = 0.001$).

Lastly, given our study design and measurement timing, we did not conduct mediation analyses for the changes in satisfaction scores. While we measured potential mediators such as psychological ownership, these variables were assessed pre-feedback and therefore could not capture the causal mechanisms driving satisfaction changes. Future research should incorporate repeated measurements of these potential mediating variables both before and after feedback to better understand the psychological mechanisms through which AI customization and reviewer characteristics influence satisfaction over time.

5.3.4. Discussions of the post-feedback results

In short, our post-feedback results further examined how AI customization and reviewer type influenced participants' responses to negative feedback on an AI-generated report. The results revealed several key insights into how these factors shape responsibility attribution and post-feedback satisfaction.

Regarding responsibility attribution, participants who customized the AI assistant exhibited different responses in how they distributed responsibility and blame for the negative feedback on the report, than those who did not customize the AI assistant. Specifically, those who customized the AI assistant took on more personal responsibility for the report's negative evaluations. While psychological ownership of the AI assistant and the report post-feedback was not measured, this finding suggests that customization leads to a greater sense of ownership and accountability, consistent with our pre-feedback findings. The type of reviewer also impacted responsibility attributions. Participants attributed more responsibility to the AI assistant when the negative feedback came from an AI reviewer than from a human non-expert, but not significantly more than from a human expert. This suggests that participants may hold the AI assistant less accountable when feedback originates from a source that they perceived as less credible. Furthermore, participants blamed the human non-expert reviewer significantly more so than both the human expert and the AI reviewer. This suggests that participants viewed AI and expert human reviewers as similarly credible sources of feedback, distinct from non-expert human reviewers.

In terms of post-feedback satisfaction, both AI customization and reviewer type played significant roles. Participants who customized their AI assistant maintained higher satisfaction with the report, compared to those who used a standard AI. This aligns with the pre-feedback findings, where customization led to higher initial satisfaction. Furthermore, participants' satisfaction decreased the least when receiving feedback from a human non-expert, compared to a human expert and an AI reviewer. This suggests that the perceived expertise of the reviewer, rather than their identity as a human or AI, may drive how participants perceived the negative evaluation and its impact on their satisfaction of the report post-feedback.

Taken together, customization primarily influenced personal accountability and engagement with the feedback, as evidenced by increased self-responsibility attribution, while generally fostering a sense of ownership that can buffer against the negative effect of satisfaction decline facing criticism. Reviewer expertise, on the other hand, plays an important role in shaping how feedback is received and processed, likely through the perceived legitimacy of the criticism itself. The finding that AI reviewers were treated similarly to human experts, rather than being dismissed as less credible, suggests that participants were willing to accept AI systems as legitimate evaluators of AI-generated content. However, it is possible that this acceptance is more contingent on perceived expertise rather than the reviewer's identity as human or AI.

6. Discussion

6.1. Summary of key findings

Our research examined how individual users interact with AI systems in collaborative tasks, focusing on how these users' ability to customize AI, the role of psychological ownership, and the impact of different types of reviewers may impact their satisfactions and responsibility attribution of the outputs. Our findings revealed that when users could customize their AI assistant, they developed stronger feelings of ownership not only toward the AI itself, but also toward the work it helped create. This sense of ownership enhanced users' satisfaction with the AI-generated work, both initially and after receiving negative feedback. We also found that the source of feedback—whether from an AI system, a human expert, or a non-expert—significantly influenced how users responded to criticism and attributed responsibility. While customization helped users maintain more positive attitudes toward their work even after criticism, the perceived expertise of the reviewer strongly shaped how seriously users took the feedback. These results highlight the complexity between technical features like customization and social factors (i.e., identity of the source of feedback) in determining the psychological outcomes of human-AI collaboration.

These findings both align with and contrast recent research on human-AI collaboration. Consistent with Chung et al. (2024), who found that customization was essential for developing psychological ownership of digital avatars, our results demonstrate that active customization of AI assistants similarly fosters ownership feelings that extend to the outputs produced. Extending this finding further, our work reveals a novel serial mediation pathway whereby ownership transfers from the AI system to its outputs, a mechanism not previously documented. Additionally, our results support Böhm et al. (2023)'s findings that people evaluate AI-generated advice based on content rather than source when unaware of its origin, our findings indicate that expertise attribution may override source effects in evaluation contexts. Interesting, our results partially contrast with Hein et al. (2024), who found AI feedback was generally perceived as less accurate and motivating than human feedback. While they observed these differences primarily for positive feedback, we found that AI reviewers were treated equivalently to human experts in negative feedback conditions when positioned as having appropriate expertise. These comparisons highlight that the psychological dynamics of human-AI collaboration are more complex than simple algorithm aversion or appreciation, depending critically on how users relate to and perceive ownership of AI.

6.2. Theoretical contributions

This research makes several theoretical contributions to the existing body of knowledge on psychological ownership, responsibility attribution, and most importantly, the interaction and relationship between human and AI. Traditionally, AI has been conceptualized and studied as an instrumental tool that automates tasks and augments human capabilities (e.g., Larson, 2021; O'Neil, 2017; Shane, 2019). While this framework was suitable for earlier AI systems focused on narrow, well-defined tasks, our findings suggest that it may provide limited insights for understanding the complex dynamics of how humans may collaborate with advanced AI systems like Large Language Models (LLMs) that engage in complex cognitive work alongside humans. The emergence of psychological ownership through customization, and its subsequent impact on how users process feedback and attribute responsibility, demonstrates that users develop deeper, more nuanced relationships with AI systems than a purely instrumental view would suggest. These relationships involve elements of trust, familiarity, and emotional investment (Pierce et al., 2001; Shu & Peck, 2011) that more closely resemble collaborative partnerships than tool utilization. This aligns with recent work examining the social and relational aspects of human-AI interaction (e.g., Delgosha & Hajiheydari, 2021; Hancock et al., 2021; Lee & See, 2004; Makridakis, 2017; Raisch & Krakowski, 2021), suggesting that productive human-AI relationships may be better understood through the lens of collaboration rather than mere utilization.

Our findings advance psychological ownership theory by revealing a distinct dual pathway through which customization influences user experiences in human-AI collaboration. The serial mediation results demonstrate that customization first builds psychological ownership of the AI assistant itself, which then transfers to ownership of its outputs, ultimately enhancing user satisfaction. This serial mediation is worth noting, as it extends beyond traditional models of psychological ownership that typically focus on a single target. The study's pre-feedback results show that simply using an AI system is not sufficient for users to feel ownership and satisfaction with its outputs—developing ownership toward the AI itself appears to be a critical intermediate step.

Moreover, our findings also reveal a complex yet “protective” role of customization in human-AI collaboration, particularly when facing criticism. The post-feedback results demonstrated that while all participants experienced decreased satisfaction after receiving negative feedback, those who had customized their AI assistant maintained relatively higher satisfaction levels. While our study design is limited in providing direct empirical evidence to test the mechanisms behind this buffering effect, we speculate that this resilience may stem from the established dual ownership of both the AI system and its outputs. Our post-feedback findings showed that users who customized their AI assistants maintained higher satisfaction levels even after receiving criticism. This buffering effect of customization adds a new dimension to research on psychological ownership in technological contexts (Jussila et al., 2015; Kirk et al., 2015). Notably, the buffering effect persisted regardless of the reviewer type, suggesting that the psychological benefits of customization may be robust across different evaluative contexts. Yet

importantly, this buffering effect may be a double-edged sword. Similar to self-serving bias in performance evaluation, strong psychological ownership through customization might lead individuals to dismiss or minimize valid criticisms, potentially hampering their ability to critically evaluate AI outputs because the AI assistant and the assistant's outputs are "theirs." This concern aligns with recent findings in decision-making errors in human-AI collaborations (Rosenbacke et al., 2024) that AI can inflate false confirmation errors in human decision-making, highlighting the potential risks of over-reliance on AI systems.

Related to the point above, our findings reveal how customization fundamentally shapes responsibility attribution patterns in human-AI collaboration. When individuals customized their AI assistants, they took more personal responsibility for the AI-generated work, even after receiving negative feedback. Rather than simply enhancing perceived control, customization appears to create a shared sense of agency where users view themselves as more accountable for outcomes. This is particularly noteworthy given the "black box" nature of LLMs (Lipton, 2018), where the underlying mechanisms of AI decision-making remain unclear to users. Despite of this, customization seems to foster a sense of joint ownership and responsibility that transcends individuals' technical understanding of the AI system.

Lastly, our findings showed that participants treated feedback from AI reviewers similarly to that from human experts (i.e., they attributed similar levels of responsibility to both sources and showing similar patterns of satisfaction decline). These findings suggest that individuals' evaluation of feedback sources may be more influenced by perceived expertise than the identity of the source (human vs. AI). They also perceived the AI reviewer to have equal expertise as the human expert reviewer. We believe this expertise equivalence finding is particularly significant given ongoing debates about AI credibility and expertise (Glikson & Woolley, 2020). Several factors may explain this equivalence. First, the sophistication of current Large Language Models, including the ChatGPT 4o-mini used in our study, represents a significant advancement in AI capabilities compared to earlier systems. Second, the study's context—where participants had limited prior knowledge of mRNA technology—may have made them more receptive to AI expertise. However, if individuals indeed view AI as equally competent as a human expert, this may raise important concerns alongside our finding on psychological ownership and customization. Specifically, individuals who develop strong ownership over an AI system they view as equally expert as humans might be particularly susceptible to overlooking the AI's limitations, further amplify biases and cognitive errors in human-AI collaborations.

In essence, while these findings demonstrate benefits of AI customization in fostering productive human-AI collaboration, it is important to highlight that individuals' strong feelings of ownership to customized AI assistant can sometimes lead us to defend the AI assistant's output through emotional investment rather than objective evaluation. Understanding this tendency is crucial for individuals and organizations to implement AI systems in ways that maximize the benefits of customization while maintaining critical thinking and evaluations. This convergence of ownership and perceived expertise may foster a dangerous form of trust—one that inadvertently creates an echo chamber, questioning least when it should question most.

6.3. Managerial implications

This research also offers several practical implications for individuals and organizations integrating AI into their content creation process and workflows. As AI transitions from a mere productivity tool to a collaborative partner in knowledge work, organizations face new challenges in fostering productive human-AI relationships while maintaining critical evaluation capabilities. The dual pathway through which customization builds psychological ownership, combined with our findings about expertise perception and responsibility attribution, suggests that organizations need a thoughtful approach to managing human-AI collaboration. Recent examples like AI-generated reviews in peer-review processes (Mollaki, 2024) and AI misuse in academic settings, where students submit AI-generated work without meaningful engagement learning (Coffey, 2024), all highlight the potential pitfalls of poorly managed human-AI relationships. The creative industry also faces such challenges. In a WSJ interview, actor Joseph Gordon-Levitt raised critical concerns about AI companies using performers' work without consent or compensation to train language models, highlighting critical questions about appropriate

boundaries and attribution in human-AI collaboration (Schwartzel, 2024). These examples underscore a crucial challenge: while psychological ownership can enhance engagement and satisfaction with AI collaboration, it must be balanced with mechanisms that promote responsible use, fair attribution, and critical evaluation. Organizations must therefore consider how to simultaneously foster productive collaboration while guarding against both the formation of validation echo chambers and the misappropriation of human contributions in AI systems.

Based on our findings, we recommend a few strategies for individuals and organizations collaborating with AI systems in their workflow. While organizations should prioritize features that allow individuals to customize their AI assistants—such as personalized naming, communication styles, and domain expertise specifications—managers must recognize that the goal extends beyond mere customization options. The aim should be to foster responsible psychological ownership that enhances collaboration while maintaining critical judgment. This could be achieved through thoughtfully designed onboarding processes that emphasize building collaborative relationships with AI systems, and providing opportunities for transparent, educational training programs (Ananny & Crawford, 2018; Mittelstadt et al., 2016; Schiff, 2022). These training programs should highlight customization features not just as personalization tools, but as ways to develop more effective collaborations with AI for better performance outputs.

Furthermore, organizations and individuals who collaborate with AI must also carefully consider how to structure evaluation processes for AI-collaborative work. Our findings suggest that individuals treat AI reviewers similarly to human experts suggest both opportunities and risks in evaluation practices. While this perceived expertise equivalence could make AI review systems an efficient complement to human evaluation, individuals and organizations should be mindful of creating balanced feedback mechanisms. Rather than fully replacing human reviewers with AI systems, organizations might benefit from implementing hybrid evaluation approaches that leverage the strengths of both. For instance, organizations could implement structured review processes that encourage individuals to seek diverse perspectives (e.g., other human experts and colleagues) beyond their customized AI assistant, helping to mitigate the risk of over-reliance on a single collaborative partnership. Similarly, individuals who collaborate with AI are encouraged to seek inputs from other experts. This is especially crucial when the individuals and organizations may lack sufficient knowledge or expertise in the focal task. Recent research by Schoenegger et al. (2024) demonstrates that consulting multiple AI systems can provide benefits similar to human “wisdom of the crowd” effects, with AI aggregates performing comparably to human crowd aggregates in forecasting tasks. This suggests that, for individuals and organizations who may not have immediate access to leverage human expertise, an implementation of multi-AI review processes may be effective in reducing user AI biases and enhance evaluation quality. Moreover, given our findings about how customization affects users’ response to criticism, organizations should consider how to deliver feedback effectively when users have developed strong psychological ownership of their AI collaborations. This might involve structured feedback processes that acknowledge the collaborative nature of the work while ensuring that critical evaluation remains objective and constructive through diverse perspectives.

Additionally, the finding that participants maintained higher satisfaction when receiving feedback from non-expert reviewers has important practical implications. In many organizational contexts, peer review and 360-degree feedback systems involve input from colleagues with varying expertise levels. Our results suggest that individuals may unconsciously discount or minimize criticism from sources they perceive as less qualified, potentially creating blind spots in professional development. While this protective mechanism may preserve self-esteem and motivation in the short term, it could lead organizations to reconsider feedback design. For instance, when implementing AI-assisted work evaluation, organizations might need to carefully frame the expertise and credibility of different feedback sources to ensure that constructive criticism is internalized appropriately, regardless of its origin. The equivalence we found between AI and human expert reviewers suggests that AI feedback systems, when positioned as having appropriate expertise, may be as effective as human experts in delivering impactful feedback that users take seriously.

Finally, we recommend individuals, organizations, and policymakers to develop frameworks for responsible collaboration that balance the benefits psychological ownership with the need for critical

evaluation. The potential risks we identified—from echo chambers of validation to inappropriate use of AI outputs—suggest that organizations should establish clear guidelines for human-AI collaboration. These guidelines should address both technical aspects (such as proper documentation of AI use) and psychological elements (such as maintaining professional judgment despite strong ownership feelings). It is important to point out that psychological ownership is a concept intertwined with emotional attachment (Shu & Peck, 2011), and while this work focuses on professional collaborative contexts, our findings about how customization fosters psychological ownership raise important considerations for the broader development of social AI agents and AI companions. As customization features are key to social AI applications designed for companionship, the potential for deep emotional attachment through psychological ownership becomes a significant concern that warrants careful consideration.

6.4. Social implications

Beyond organizational contexts, our findings also suggest important implications for how society navigates the integration of generative AI into creative and intellectual work, as well as mental wellbeing. Our results indicate that psychological ownership may act as a protective buffer (Avey et al., 2009; Brown et al., 2014) and make individuals defend their AI's output. While customization may enhance engagement and satisfaction, it may also create an echo chamber that validate rather than challenge our work, and potentially amplify existing biases (Glickman & Sharot, 2025). This has concerning implications for our society.

First and perhaps most importantly, humans are prone to confirmation bias (Nickerson, 1998) and prefer validation over criticism (Hart et al., 2009). Similarly, AI systems are shaped by inherent biases from multiple sources, such as training data (Leavy et al., 2020), algorithmic designs (Ferrara, 2023), reinforcement learning from human feedback (Casper et al., 2023; Chakraborty et al., 2024). As individuals increasingly customize AI to align with their perspectives, they may co-create a closed loop based on both humans and AI systems' existing biases. The psychological risks of such validation loops extend beyond mere confirmation bias. Recent research has begun exploring whether AI interactions might trigger psychosis in vulnerable individuals, with AI systems potentially reinforcing delusional or paranoid beliefs through feedback loops (Dohnány et al., 2025; Fieldhouse, 2025). While our study focused on professional collaboration rather than clinical populations, the mechanisms we observed suggests a similar user-AI feedback loop. The high credibility attributed to AI reviewers in our study further amplifies this risk, as this perception may lead individuals to believe their biases are supported and endorsed by credible experts. This could undermine not only the social processes through which knowledge is contested and refined, but also individual psychological wellbeing. Educational institutions, professional communities, and mental health practitioners may need to develop new frameworks for recognizing and addressing these risks. Proactive measures could include encouraging individuals to engage with AI-generated outputs critically, providing targeted support systems for vulnerable populations (e.g., individuals already experiencing cognitive or emotional distress).

As mentioned earlier, our finding suggest that AI reviewers are perceived as equivalent to human experts in evaluative contexts. This equivalence has important social implications. As AI systems increasingly participate in peer review, educational assessment, and creative evaluation, the tendency to grant AI systems expert-equivalent status could accelerate the legitimization of AI judgment in domains previously reserved for human expertise. On one hand, this may accelerate productivity across various domains. On the other hand, this may severely impact the development of human critical thinking and creativity. If individuals defer too readily to AI judgments, they may engage less deeply with the evaluative process, limiting opportunities for critical reflection and human originality. Over time, this deference could lead to a diminishment of collective diversity of creative ideas, as documented in recent studies (Meincke et al., 2025; Moon et al., 2025).

In sum, we propose that the path forward may require careful and deliberate social choices about how AI is integrated into human intellectual and productive activities. Our findings suggest a need for policymakers, educators, and professional communities to establish frameworks that safeguard critical discourse, ensure authentic human contribution in human-AI collaborative activities. Such frameworks may include guidelines for transparency in AI-assisted evaluation, mechanisms to preserve human

inputs and accountability in decision-making, as well as policies that prevent the concentration of AI benefits among limited groups.

6.5. Limitations and future directions

While our research provides important insights into psychological ownership and human-AI collaboration, several limitations and opportunities for future research warrant discussion. First, our methodological choices, while necessary for experimental control, present certain limitations. Our design would have benefited from measuring psychological ownership at both pre- and post-feedback time points, which would enable more comprehensive analysis of how ownership feelings evolve in response to feedback. Future research should consider incorporating both pre- and post-feedback mechanism measures while effectively balancing this with participant fatigue.

Our use of ChatGPT 4o-mini, while allowing consistent access and interactive experience across participants, raises questions about generalizability to other AI systems. Additionally, given that ChatGPT 4o-mini utilized in this study has a comparatively smaller parameter set and more restricted training data access than some more recently developed models (Ben Abacha et al., 2024), the finding that participants rated it as almost equivalent to human experts in a cutting-edge, biotechnological domain suggests the need to examine how different AI capabilities might affect perceived expertise and psychological ownership. Future research should explore these dynamics across various AI technologies, from more sophisticated large language models to other forms of generative AI and even industrial robotics.

Beyond methodological considerations, our findings highlight several conceptual areas requiring further investigation. One key area concerns the nature of intimate knowledge in human-AI collaboration. While psychological ownership theory suggests that intimate knowledge of a target strengthens ownership feelings, AI systems present a unique challenge: individuals may develop familiarity with an AI's "personality" and task-specific capabilities without understanding its underlying mechanisms. This black box nature of AI (Lipton, 2018; Ribeiro et al., 2016) raises important questions about how different types of knowledge—surface-level vs. algorithmic understanding—might affect psychological ownership and collaboration patterns in different ways. Future research could also measure the three antecedents of psychological ownership and examine how each of these antecedents impact the facilitation of psychological ownership and their downstream consequences.

The type and depth of customization also warrants further examination. Our study suggests that customization fosters psychological ownership, but future research should investigate whether different forms of customization (e.g., superficial naming versus core functionality modifications) produce varying levels of ownership and responsibility attribution. Similarly, individuals' preexisting attitudes toward AI algorithms may moderate these relationships. For example, those with high algorithmic appreciation and are ready to adopt to new technologies (Logg et al., 2019) might be more likely to develop ownership feelings toward their AI assistants and AI-generated outputs, while those with algorithmic aversion might remain skeptical regardless of customization opportunities.

Given the growing size and research interests of shared economies, it may be highly interesting to examine human-AI collaborations in team settings. Drawing parallels from research on shared economy and collective psychological ownership (Lee et al., 2019; Morewedge et al., 2021; Pierce & Jussila, 2010), future studies could investigate how teams manage customized AI preferences, resolve conflicts over AI usage, and develop shared norms for human-AI collaboration. For instance, when working on the same project, different team members may have their own preferences with collaborating with AI and how to customize AI for collective outcomes. Some team members might develop strong psychological ownership through customization while others prefer different settings or approaches, potentially leading to conflicts over AI configuration and usage patterns. Moreover, the question of how collective psychological ownership develops in team-AI collaborations remains unexplored. Teams might need to balance individual members' customization preferences with the need for consistent AI interaction patterns across the group. Furthermore, understanding how teams attribute responsibility for AI-generated outcomes when multiple members have contributed to customizing and directing the AI

could provide valuable insights for managing collective human-AI collaboration processes and outcomes.

Future research could explore how humans develop different psychological relationships with different instances (i.e., chat sessions) of the same AI model. Even when individuals are aware they are interacting with the same large language model, they may attribute distinct personalities, responsibilities, or capabilities to different sessions or instances. This raises interesting questions about the nature of psychological ownership and responsibility attribution in human-AI collaboration: do individuals develop psychological ownership of an AI model itself, or do individuals develop different relationships with different instances of the same AI? And how would this affect their sense of ownership, responsibility attributions, and other attitudes and behaviors when collaborating with AI? These research questions may provide valuable insights into the psychological mechanisms underlying human-AI relationships and inform the design of future collaborative AI systems.

Notes

1. Significant covariates included participants' general tendency to customize AI ($p < 0.001$), understanding of AI algorithms ($p = 0.02$), knowledge of mRNA vaccines ($p < 0.001$), and age ($p = 0.01$).
2. Significant covariates included participants' general tendency to customize AI ($p = 0.01$), knowledge of mRNA vaccines ($p < 0.001$), and age ($p = 0.04$).
3. Significant covariates included participants' age ($p < 0.001$) and knowledge of mRNA vaccines ($p = 0.003$).
4. Participants' self-reported knowledge of mRNA vaccines was a significant covariate ($p = 0.001$).
5. Participants' self-reported knowledge of mRNA vaccines was a significant covariate ($p < 0.001$).
6. Significant covariates included participants' age ($p = 0.04$), general tendency to customize AI ($p = 0.03$), and understanding of AI algorithms ($p = 0.02$). Additionally, participants' perceived reviewer expertise significantly interacted with time ($F(1, 495) = 48.95$, $p < 0.001$), suggesting that the impact of negative feedback on satisfaction varied depending on how expert participants perceived the reviewer to be.
7. Participants' self-reported knowledge of mRNA vaccines was a significant covariate ($p < 0.001$).

Author contributions

CRedit: **Yunzhijun Yu**: Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Writing – original draft, Writing – review & editing; **Mehmet A. Yetim**: Conceptualization, Data curation, Writing – review & editing; **Ovidiu Cocieru**: Conceptualization, Writing – review & editing.

Declaration of generative AI and AI-assisted technologies in the writing process

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Data availability statement

The data supporting the findings of this study are available at <https://osf.io/a92cq>.

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